

Notes: Dyck, Morse & Zingales (JF, 2010)

Who Blows the Whistle on Corporate Fraud?

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1 Big picture

- **Question.** Who actually detects large corporate frauds, and what incentives make them speak up?
- **Setting.** The paper studies 216 alleged corporate frauds at large U.S. companies between 1996 and 2004, after screening for frivolous securities class-action suits.
- **Main finding.** Fraud detection does not rely mainly on standard governance actors such as the SEC, auditors, boards, or investors. It “takes a village,” including employees, media, industry regulators, analysts, short sellers, clients, suppliers, and others.
- **Surprising actors.** Employees detect roughly 17% of external detections; media and industry regulators also play large roles. Traditional legal and finance views understate these actors.
- **Two missing dimensions.** The paper argues that standard theories miss (i) access to information and (ii) nontraditional incentives such as reputation, career concerns, and monetary whistleblower rewards.
- **Market reaction.** The disclosure of fraud is economically important: the event study shows a sharp negative abnormal return at the date of fraud revelation.
- **Incentive evidence.** Monetary incentives help explain employee whistleblowing, especially in health care where qui tam rewards exist. Reputational incentives are weak for auditors and analysts but stronger for journalists in large cases.
- **Policy implication.** A cost-effective fraud-detection system should not focus only on classic gatekeepers. It should also protect and incentivize employees and information intermediaries who already have access to relevant information.

2 Incremental contribution of the paper

- **First broad empirical map of fraud detection.** Rather than studying one detector type, the paper constructs a comprehensive taxonomy of who first reveals corporate fraud.
- **Moves beyond formal governance.** It shows that informal and nontraditional actors are central to detection, challenging legal, private litigation, and finance views of corporate governance.
- **Separates access from incentives.** The paper distinguishes actors who possess information as a byproduct of their job from actors who have financial or reputational incentives but high information-gathering costs.
- **Links whistleblowing to incentives.** The paper studies specific incentive mechanisms: auditor reputation, analyst career advancement, journalist career gains, employee costs and rewards, health-care qui tam rewards, and SOX changes.
- **Quantifies private costs and benefits.** The employee table documents retaliation, job loss, legal exposure, and financial benefits, making whistleblowing a cost-benefit problem rather than a moral anecdote.
- **Policy relevance.** The paper evaluates whether reforms such as SOX affected detection patterns and suggests that stronger monetary incentives and retaliation protection may matter more than simply increasing formal gatekeeper duties.

Interpretation: The key contribution is to recast fraud detection as an ecosystem problem. Detection depends not only on who should monitor, but also on who has access to information, who has enough incentive to reveal it, and who can survive the personal costs of doing so.

3 Data and measurement

3.1 Fraud sample

The sample consists of alleged corporate frauds in U.S. companies with more than \$750 million in assets between 1996 and 2004. The authors start from securities class-action filings and then screen out frivolous cases, leaving 216 fraud cases.

Interpretation: The sample focuses on large firms because these are the cases where standard governance mechanisms should be most active. If formal gatekeepers fail here, the finding is especially important.

3.2 Detector coding

The core variable is the *detector of fraud*: the actor first credited with identifying or revealing the fraud. Detector categories include internal governance, analysts, auditors, clients/competitors, employees, equity holders, industry regulators, law firms, media, SEC, short sellers, and others.

Interpretation: This coding is the heart of the paper. The contribution is not merely counting SEC cases or auditor cases; it reconstructs each fraud’s detection history to identify who made the fraud public.

3.3 Adjustments for short sellers and case value

The paper adjusts the raw distribution in two ways:

- It reclassifies cases as short-seller detections when short interest prior to revelation is unusually high.
- It value-weights cases using settlements and fines to account for the economic size of the fraud.

Interpretation: The short-selling adjustment matters because short sellers may detect fraud without appearing in the public revelation. Value weighting matters because small and large frauds may involve different detectors.

3.4 Incentive variables

The paper constructs variables that capture potential incentives and information costs:

- legal view, private litigation, finance view, access to internal information, external access, career concerns, monetary rewards, and financial risk;
- auditor turnover and auditor reputation variables;
- analyst All-Star status, investment-bank ranking, and analyst tenure;
- journalist promotion and demotion outcomes;
- employee costs, lawsuits, retaliation, and monetary awards;
- health-care industry and qui tam monetary incentives.

Interpretation: These variables allow the paper to ask why some actors speak while others stay silent. Detection is modeled as a function of expected private benefits, costs, and access to information.

4 Models and empirical specifications

4.1 Distribution of detectors

The first object is a descriptive distribution:

$$Share_k = \frac{\#\{\text{frauds detected by actor } k\}}{\#\{\text{frauds}\}}.$$

The authors report equal-weighted, short-seller-adjusted, value-weighted, and smoking-gun-only versions.

Interpretation: The distribution reveals whether existing theories focus on the right actors.

4.2 Conditional logit model of detector type

For each fraud case, the paper constructs a choice set of possible detector types. A simplified representation is:

$$Pr(Detector_{ic} = k) = \frac{\exp(X'_{ik}\beta)}{\sum_j \exp(X'_{ij}\beta)},$$

with case fixed effects. The covariates represent detector-type theories and incentives.

Interpretation: The model asks which characteristics make a detector type more likely to be the one that reveals a given fraud.

4.3 Auditor choice after turnover

The auditor analysis asks whether firms leaving an auditor choose new auditors associated with lower fraud risk:

$$Pr(NewAuditor_{if} = a) = \Lambda(\alpha + \beta FraudShare_a + \gamma Controls_a).$$

Interpretation: If auditor reputation is strong, firms should avoid auditors associated with prior frauds. The paper finds evidence that firms do avoid such auditors.

4.4 Analyst and journalist career regressions

The career tests compare whistleblowers with comparable non-whistleblowers:

$$CareerMove_i = \alpha + \beta Whistleblower_i + \gamma Controls_i + \varepsilon_i.$$

Outcomes include analyst promotion/demotion and journalist movement to higher-status jobs.

Interpretation: These regressions test reputational incentives. If whistleblowing is rewarded, whistleblowers should have better career outcomes than matched peers.

4.5 Employee monetary incentive test

The employee test compares health-care and non-health-care frauds, because government health-care spending creates qui tam reward possibilities:

$$Pr(EmployeeDetector_i = 1) = \Lambda(\alpha + \beta HealthCare_i + \gamma Controls_i).$$

Interpretation: The health-care comparison is the paper’s main test of monetary incentives for employees. A positive coefficient implies that employees reveal more when monetary rewards are available.

5 Identification and empirical design

5.1 Nature of identification

The paper is not a randomized experiment. Identification comes from combining a comprehensive sample, careful case histories, theory-based detector characteristics, and targeted incentive tests. The empirical design is best described as comparative institutional evidence.

Interpretation: The strongest claim is descriptive and mechanism-based: different actors detect fraud at different rates because they face different information costs and incentives.

5.2 Why the “who detects” distribution matters

The first empirical object is descriptive but fundamental. If auditors, SEC, law firms, and equity holders rarely detect fraud, then theories and regulations centered only on these actors miss important parts of the fraud-detection process.

5.3 Testing competing theories

The conditional logit tests legal, private litigation, finance, information-access, monetary, and career-concern explanations in one framework. This lets the authors compare theories rather than report isolated case narratives.

5.4 Incentive-specific designs

- **Auditors:** Do clients avoid auditors associated with past frauds?
- **Analysts:** Do analysts who reveal fraud move up in rankings or investment-bank status?
- **Journalists:** Do reporters who break fraud stories advance relative to matched reporters?
- **Employees:** Are employee detections more common where qui tam monetary rewards are possible?
- **SOX:** Did the detection mix change after Sarbanes-Oxley?

Interpretation: These designs convert broad incentive arguments into measurable tests. The evidence is strongest for monetary incentives among employees and career incentives among journalists.

5.5 Main limitations

- The sample contains revealed frauds, not all frauds.
- Detector coding requires judgment and reading of case histories.
- Some actors may contribute behind the scenes but not receive credit as detectors.
- The incentive tests are observational and cannot fully rule out unobserved differences across industries or actors.
- The results apply to large U.S. frauds in 1996–2004, a period surrounding major scandals and SOX.

6 Tables and figures

6.1 Table I: Data Definition and Sources; Figure 1: Event study around the whistleblowing date

Read: Table I defines the paper’s core variables and data sources. Figure 1 shows the stock-market reaction around fraud revelation.

- The event study displays a sharp negative abnormal return on the revelation date.
- Panel A uses OLS estimates; Panel B uses median regressions.
- The drop is economically large, confirming that detection events reveal valuable negative information.

Economic message: Fraud detection is economically important. The event study justifies why it matters who detects fraud and how incentives can improve detection.

Data Definition and Sources		
This table describes the main variables used in our analysis and provides the sources.		
Variable	Description	Sources
Detector of fraud	The actor to first identify the fraud based on our reading of the legal case and an average of 800 articles from Factiva in a window from 3 months before the class action period to settlement. Detector categories include: auditor, analyst, equity holder, short seller, media, clients & suppliers, financial market regulators, non-financial-market regulators, employees, and lawyers. Media is credited only when the story does not indicate another actor as the source. Financial market regulators are the SEC and stock exchanges. Non-financial regulators include industry regulators (e.g., FERC, FAA, FDA) and government agencies.	Security Class Action filings available from Stanford Securities Class Action Clearinghouse (SSCAC) database, Factiva articles.
Settlements and fines	The sum of the settlement amount paid to shareholders in the class action lawsuit; any fines or settlements paid to the SEC, or the courts by the firm, its insurance, or its officer/directors; and any fines or settlements paid to the courts or regulators by the firm’s agents (auditors and investment banks) regarding the impropriety.	Security Class Action filings in SSCAC database, SEC, Factiva articles.
Assets	The dollar value of assets in the year prior to the revelation of the fraud.	Compustat.
Fraud duration	The class period defined in the final court-certified security class action suit. We restrict the maximum duration to 3 years, to avoid changes in duration possibly arising from changing rules with the passage of SOX in July of 2002.	SSCAC.
Financial restatement dummy	Value of 1 given for filing a 10-Q/A or 10-K/A or an 8-K that referred to restatement information (116 cases); announcing an intention to restate the firm’s financials but not as a result of bankruptcy (e.g., Enron) (7 cases); taking a one-time accounting-related charge (6 cases); or having accounting-related investigations for ongoing cases (3 cases).	SEC filings, General Accounting Office (GAO) report on Financial Statement Restatements.

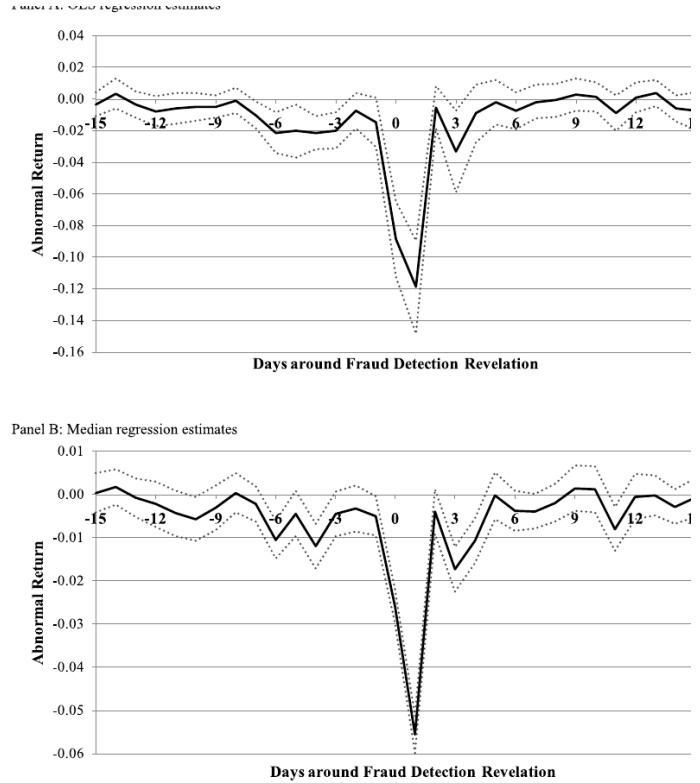


Figure 1. Event study around the whistleblowing date. For each company, we consider the time interval from the beginning of the class action period (the date when the fraud alleged began) until 1 year after the end of the class action period. For each firm, we regress stock return on the S&P500 returns and an indicator variable marking the date of the news article that reports on the fraud detection. We then repeat the regression 30 times, changing the event date in each of the regressions such that we run the single-day event study for all dates between 15 days before and 15 days after the news article date. Figure 1, Panel A reports the estimates of the event dummy using an OLS regression, while Figure 1, Panel B reports the estimates using a median regression.

(a) Table I: Data definitions and sources.

(b) Figure 1: event study around fraud revelation.

The table defines the measurement system; the figure shows why detection is valuable.

6.2 Table II: Who Detects Corporate Fraud?; Table III: Competing Theories of Who Blows the Whistle - Results

Read: Table II reports the distribution of fraud detectors. Table III estimates conditional logit models of detector choice.

- External governance actors reveal most frauds.
- Employees, media, industry regulators, analysts, and short sellers are all important.
- Monetary rewards and career concerns help explain detection.
- External access costs matter, especially in equal-weighted results.

Economic message: Fraud detection is not concentrated in one institutional actor. The system works as a decentralized network of information holders and incentive bearers.

TABLE II
Who Detects Corporate Fraud?

We identify a case as one of *internal* governance when the revealer of fraud is firm manager (e.g., via a press release or resignation) or the board of directors. Column 1 is the original coding. In column 2, we recode the fraud detector as a short seller when short selling activity prior to revelation is more than three standard deviations above the prior 3-month average. Column 3 adjusts column 2 to reflect a value-weighting of cases, where the weights are the adjusted value of the sum of settlements and fines. The adjustment is the winsorized settlement value. For the few cases that have not settled or where the settlement amount was not made public, we use the median settlement amount. The final column presents, for robustness, the cases that we identify as a "smoking gun." See the Internet Appendix for further information on reclassification settlements and fines and smoking gun classifications.

Panel A: Distribution of Fraud Detectors				
Fraud Detector	Raw Distribution (Equal Weight)	Data Adjusted for Short Activity (Equal Weight)	Data Adjusted for Short Activity (Value Weight)	Robustness Smoking Gun Only (Equal Weight)
	(1)	(2)	(3)	(4)
Internal governance	74 (34.3%)	64 (29.6%)	60 (27.9%)	n/a
External governance	142 (65.7%)	152 (70.4%)	156 (72.1%)	112
Total cases	216 (100%)	216 (100%)	216 (100%)	112 (100%)
Fraud Detectors within External Governance				
Analyst	24 (16.9%)	21 (13.8%)	24.1 (15.9%)	18 (16.1%)
Auditor	16 (11.3%)	16 (10.5%)	11.3 (7.4%)	13 (11.6%)
Client or competitor	9 (6.3%)	7 (4.6%)	2.7 (1.8%)	4 (3.6%)
Employee	26 (18.3%)	26 (17.1%)	25.6 (16.8%)	21 (18.8%)
Equity holder	5 (3.5%)	5 (3.3%)	5.3 (3.5%)	5 (4.5%)
Industry regulator, govt. agency or self-regulatory organization	20 (14.1%)	20 (13.2%)	14.1 (9.3%)	17 (15.2%)
Law firm	5 (3.5%)	5 (3.3%)	3.5 (2.3%)	2 (1.8%)
Media (incl. academic publications)	22 (15.5%)	20 (13.2%)	35.7 (23.5%)	13 (11.6%)
SEC	10 (7.0%)	10 (6.6%)	8.6 (5.7%)	8 (7.1%)
Short-seller	5 (3.5%)	22 (14.5%)	21.2 (13.9%)	11 (9.8%)
External governance total cases	142 (100%)	152 (100%)	152 (100%)	112 (100%)

Competing Theories of Who Blows the Whistle—Results

The table reports conditional logit tests of fraud detection theories, where the dependent variable is an indicator for being the fraud detector for that case with case fixed effects. Panel A reports results for equal-weighted tests, and Panel B value-weighted test results. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The mfx column reports marginal effects for significant variables in the previous column.

Theories	Dependent Variable: Choice of Fraud Detector among 10 Types					
	(1)	(2)	(3) mfx	(4)	(5)	(6)
Panel A: Equal Weighted						
Legal view	−0.962*** (0.308)					
Private litigation	−1.460*** (0.470)					
Financial risk	−0.183 (0.213)	−0.167 (0.216)		−0.268 (0.232)	−0.290 (0.241)	−0.315 (0.244)
Monetary rewards	1.065*** (0.290)	0.949*** (0.273)	0.230	0.934*** (0.280)	0.937*** (0.280)	0.933*** (0.281)
Career concerns	1.047*** (0.249)	0.464** (0.201)	0.115	0.453** (0.211)	0.453** (0.213)	0.451** (0.214)
External access	−0.953*** (0.239)	−0.596*** (0.205)	−0.148	−0.671*** (0.214)	−0.685*** (0.221)	−0.700*** (0.221)
All-Star analyst %				1.374** (0.681)		1.165 (0.871)
Analyst tenure					0.062* (0.037)	0.026 (0.049)
Observations	1,520	1,520		1,480	1,480	1,480
Pseudo R ²	0.059	0.027		0.031	0.029	0.031
Panel B: Value Weighted						
Legal view	−1.529*** (0.539)					
Private litigation	−1.993*** (0.676)					
Financial risk	−0.172 (0.346)	−0.169 (0.369)		−0.282 (0.426)	−0.503 (0.456)	−0.503 (0.459)
Monetary rewards	1.412*** (0.445)	1.203*** (0.413)	0.266	1.211*** (0.418)	1.210*** (0.424)	1.210*** (0.424)
Career concerns	1.573*** (0.403)	0.741** (0.324)	0.183	0.748** (0.337)	0.751** (0.358)	0.751** (0.358)
External access	−1.097*** (0.403)	−0.456 (0.354)		−0.521 (0.381)	−0.659* (0.393)	−0.659* (0.394)
All-Star analyst %				1.033 (1.032)		−0.043 (1.444)
Analyst tenure					0.102 (0.066)	0.104 (0.083)
Observations	1,520	1,520		1,480	1,480	1,480
Pseudo R ²	0.097	0.033		0.035	0.040	0.040

(a) Table II: distribution of fraud detectors.

(b) Table III: competing theories in conditional logit models.

The main descriptive and explanatory tables for who detects corporate fraud.

6.3 Table IV: Auditor Turnover; Table V: Do Analysts Who Blow the Whistle Advance Their Careers?

Read: Table IV studies auditor reputation through auditor turnover and new-auditor choice. Table V studies whether analysts benefit from whistleblowing.

- Auditor whistleblowing firms have high auditor turnover.

- Firms choosing new auditors appear to avoid auditors with higher prior fraud exposure.
- Analysts who blow the whistle do not show clear career advancement.

Economic message: Auditor reputation matters on the client side, but analyst reputational rewards appear weak. This helps explain why analysts detect some frauds but not as much as the finance view might predict.

Auditor Turnover

The first row of Panel A reports the turnover of auditors in the sample of all large firms (more than \$750 million in assets) during the sample period, excluding Arthur Andersen clients during 2001 to 2002. The next two rows report the turnover of auditors for the fraud firms, separately for the fraud being revealed by internal (second row) and external (third row) mechanisms. The fourth row reports the turnover of auditors in the subset of firms where the whistleblower was the auditing firm. The third column of Panel A reports the p -value from a simple binomial test that the probability of turnover is different from the all-firms frequency in row 1. Panel B reports summary statistics for the independent variables. Panel C presents the results of a conditional logit assumption of the choice of new auditor following auditor turnover. The first two columns limit the analysis to turnovers in 1999 and 2000. Columns 3 and 4 limit the sample to the selection of audit in 2002 for clients of Arthur Andersen as of 2000. The columns labeled mfx report the margin effects from the prior column for the significant coefficients. Standard errors are in parentheses. ***, **, and * indicate significant differences at the 1%, 5%, and 10% levels, respectively.

Panel A					
Turnover in	Frequency	Observations	p -Value for Difference from Row 1		
All large firms 1996–2004	0.052	20,171			
Fraud firms, internal 1996–2004	0.141	64	0.006		
Fraud firms, external 1996–2004	0.147	136	0.000		
Auditor whistleblowing firms	0.500	16	0.000		
Panel B					
3 Prior Year Cumulates of	1999 & 2000 Turnovers, Sample of 290 Auditor Changes		2002 Arthur Andersen Turnovers, Sample of 356 Auditor Changes		
	Mean	Std. Dev.	Mean	Std. Dev.	
Frauds in auditing firm clients	7.00	3.58	26.3	5.66	
Accounting frauds in clients	4.52	2.25	15.7	3.73	
Whistleblowing by auditor	–	–	0.913	0.250	
Frauds/market share	39.4	20.4	110.9	56.7	
Accounting frauds/market share	25.7	13.7	66.0	34.8	
Whistleblowing/market share	–	–	3.99	3.92	
Panel C					
Conditional Logit Estimation	Choice of New Auditor Turnovers 1999 & 2000		Arthur Andersen Clients' Auditor Choice in 2002		mfx
	(1)	mfx (2)	(3)	mfx (4)	
Proportion frauds	–0.016** (0.006)	–0.0033 (0.006)	–0.011* (0.001)	–0.0024 (0.001)	–0.0022** (0.011)
Proportion accounting frauds		–0.012* (0.007)	–0.0026 (0.007)	0.024 (0.020)	
Proportion frauds where whistleblower				0.033 (0.041)	
Big 5/4	2.10*** (0.331)	2.28*** (0.338)	4.47*** (0.515)	4.61*** (0.523)	
Observations	2,320	2,320	2,208	2,208	
Decomposed R^2	0.079	0.076	0.091	0.097	

(a) Table IV: auditor turnover.

Reputational incentives are actor-specific: auditors face firm-choice consequences, while analyst career benefits are limited.

Do Analysts Who Blow the Whistle Advance Their Careers?

This table provides statistics and tests for differences in the promotion and demotion probabilities between sell-side equity analysts who blow the whistle and analysts in the firms where a whistle was blown that do not themselves blow the whistle. All-Star rankings are derived from *Institutional Investor* rankings. Ranking of investment banks follows the classification in Hong and Kubi (2003), applied to our sample period. See Table I for further information. Analyst information from I/B/E/S. In Panel C, columns 1 and 2, the dependent variable takes the value one if the analyst became an All-Star following the whistle being blown (and was not beforehand), and zero otherwise. In Panel C, columns 3 and 4, the dependent variable takes the value one if the analyst loses an All-Star ranking following the whistle being blown. Regressions include company fixed effect. Robust standard errors are presented in parentheses. ***, **, and * indicate significant differences at the 1% and 5% levels, respectively.

Panel A: % of Highly Ranked Analysts among All I/B/E/S Analysis Covering Fraud-Committing Firms				
	Whistleblower	Non-whistleblower	<i>p</i> -Value (Diff)	
Pre-period All-Star analyst	50%	9.8%	0.000***	
Pre-period employed at high-tier I-bank	60%	38%	0.053*	
Observations	20	397		
Panel B: Career Advancement of I/B/E/S Analysts Covering Fraud-Committing Firms				
		whistleblower	Non-whistleblower	<i>p</i> -Value (Diff)
All-Star analyst				
Promoted to All-Star in	1 year	10.0%	4.5%	0.41*
	2 years	12.5%	5.4%	0.39*
Demoted from All-Star in	1 year	20.0%	18.4%	0.91*
	2 years	22.2%	50.0%	0.13*
I-bank ranking				
Promoted to high-tier I-bank in	1 year	0	1.0%	0.78*
	2 years	0	3.8%	0.60*
Demoted from high-tier I-bank in	1 year	0	4.3%	0.46*
	2 years	0	8.5%	0.33*
Panel C: Logit Test of Advancement Difference for All-Star Analysts				
Dependent Variable	Promoted		Demoted	
	One Year	Two Years	One Year	Two Years
Whistleblower	0.921 (1.540)	0.871 (1.522)	0.618 (1.653)	−2.562** (1.286)
Experience	0.528 (0.395)	0.920** (0.411)	−0.630 (0.681)	0.030 (0.525)
Pseudo <i>R</i> ²	195	155	20	34
Observations	0.104	0.173	0.149	0.155

(b) Table V: analyst career outcomes.

6.4 Table VI: Who in the Media Detects Fraud?; Table VII: Does Whistleblowing Advance Journalists' Careers?

Read: Table VI lists the media outlets and journalists associated with fraud revelations. Table VII tests journalist career outcomes.

- Prominent newspapers such as the New York Times and Wall Street Journal appear repeatedly.
- Many fraud stories are associated with named journalists.
- Journalists who reveal fraud show improved career movement relative to matched non-whistleblowers, especially over three years.

Economic message: Media whistleblowing is supported by reputational rewards. This is one reason the media becomes more important in value-weighted frauds.

Who in the Media Detects Fraud?

For each case in which the media is the fraud detector, the table records the newspaper or journal that reveals the fraud, the reporter(s) of the article, and the page on which the article appears.

Company	News Outlet	Reporter	Article Location
AOL TimeWarner	<i>New York Times</i>	Gretchen Morgenson	Page 1, Business
Computer Associates	<i>New York Times</i>	Alex Berenson	Page 1, Business
Halliburton	<i>New York Times</i>	Alex Berenson and Lowell Bergaman	Page 1, Business
Sprint	<i>New York Times</i>	David Cay Johnston	Page 25, Section 1
Ascend	<i>San Francisco Chronicle</i>	Herb Greenberg	Page 1, Business
Communications	<i>Wall Street Journal</i>	Molly Williams	Page C11, Heard on the Street
Broadcom	<i>Wall Street Journal</i>	Jonathan Weil	Page C1, Heard on the Street
Cardinal Health	<i>Wall Street Journal</i>	Jonathan Weil	Page T1—regional front page of WSJ
Enron	<i>Wall Street Journal</i>	Jonathan Weil	Page A10
E.W. Blanch	<i>Wall Street Journal</i>	Deborah Lohse	Pages A1, B6
Qwest	<i>Wall Street Journal</i>	Deborah Solomon, Steve Liesman, Denis Berman	
Raytheon	<i>Wall Street Journal</i>	N/A	
AT&T	<i>Business Week</i>	Robert Barker	Investor column (p. 264)
Bausch & Lomb	<i>Business Week</i>	Rochelle Sharpe	Page 87
Silicon Graphics	<i>Business Week</i>	Robert Hof, Ira Sager, Linda Himelstein	Cover story
Apria Health Care	<i>Fortune</i>	Erick Schonfeld	Page 114
Sunbeam	<i>Barrons</i>	Jonathan Laing	Page 17
Cambrex	<i>Chemical Reporter</i>	N/A	N/A
Long Island Lighting	<i>Daily Electricity Reporter</i>	N/A	N/A
Bristol Myers Squibb	<i>Cancer Letter</i>	N/A	N/A

(a) Table VI: media fraud detections.

Does Whistleblowing Advance Journalists' Careers?

This table provides statistics and tests for differences in the promotion and demotion probabilities between reporters who blow the whistle identified in Table VII and reporters with a similar status at the same time in the same media outlet who did not blow the whistle. See Table I for further details about peer construction. Panel A reports the movement distribution, where movement is categorized as moving to a lower job, staying in the same job or equivalent job, or moving to higher job. Panel B tests whether the mean movement is different for the whistleblower and non-whistleblower samples. An F -test is used to allow for weighting of the peers such that there is or peer and one whistleblower for each case. ** and * indicate significant differences at the 5% and 10% levels, respectively.

	One Year Post-Fraud		Three Years Post-Fraud	
	Whistleblower	Non-whistleblower	Whistleblower	Non-whistleblower
Panel A: Distribution of Career Promotions & Demotions				
Lower job	0 (0%)	18 (12%)	1 (6%)	39 (26%)
Equivalent job	14 (82%)	120 (78%)	12 (71%)	80 (53%)
Higher job	3 (18%)	16 (10%)	33 (24%)	33 (22%)
Panel B: Test for Difference in Mean Movement (Mean Movement Is Coded +1 = Promoted, 0 = No Change, -1 = Demoted)				
Mean movement	0.153	-0.086	0.289	-0.083
	Ho: Whistle - Non-whistle = 0		Ho: Whistle - Non-whistle = 0	
	$F(1, 167) = 2.75^*$		$F(1, 167) = 3.99^{**}$	
	Prob > $F = 0.0990$		Prob > $F = 0.0475$	

(b) Table VII: journalist career outcomes.

Media detection is tied to outlet credibility and reporter career incentives.

6.5 Table VIII: What Are the Costs and Benefits for Employees to Whistleblow?

Read: Table VIII gives detailed case-level evidence on employee whistleblowers.

- Many employees experience termination, reduced responsibility, litigation, retaliation, or career loss.
- Some receive legal or monetary benefits, especially under whistleblower statutes.
- The costs are concrete and often severe, explaining why employee whistleblowing may require strong protection or reward mechanisms.

Economic message: Employees are well informed but face high personal costs. Without monetary rewards or anti-retaliation protection, many will rationally remain silent.

What Are the Costs and Benefits for Employees to Whistleblow?

The table indicates for each employee whistleblower the following information: company (column 1); the whistleblower's name and position (column 2); whether the whistleblower was terminated, quit, or given a job with significantly reduced responsibility (column 3); other costs claimed by the employee (column 4); whether a lawsuit was filed with potential for damages, including the type of lawsuit (column 5); whether an outcome to the lawsuit if any (column 6); and other possible benefits of whistleblowing (column 7). Panel A reports results for whistleblowers where the name of the whistleblower was revealed. Panel B reports results for whistleblowers that remain unnamed.

Company (1)	Whistleblower, Position (2)	Costs		Benefits		
		Terminated, Quit, or Reduced Responsibility? (3)	Other Costs (4)	Filed Lawsuit with Potential for Damages (5)	Positive Outcome of Lawsuit (6)	Other Possible Benefits (7)
		Panel A: Named Whistleblowers				
Apria Health Care	Mark Parker, Branch Manager	Yes		Yes— <i>qui tam</i> , wrongful dismissal	No—government doesn't join	Retaliation
Citizens Utilities	Robert Arnold, Project Manager	Yes		Not clear—state filed lawsuit, gets lower rates		
Columbia HCA Health Care/ Osten (2 cases)	Donald McLendon, executive of acquired firm	Yes	Could not find other job, financial stress.	Yes— <i>qui tam</i>	Yes—\$35 million	Avoid potential legal liability
Dynegy	Ted Beatty, Management Trainee	Yes	Could not find other job, forced to leave hometown, home broken into, threats and intimidation.	No		Retaliation
Endocare	Joseph Hafemann, Corporate Controller	Yes		No		Avoid potential legal liability
GTECH holdings	David Armitage, <i>Entrepreneur</i>	No		No		Vengeance

(a) Table VIII, part 1.

Table VIII—Continued

Company (1)	Whistleblower, Position (2)	Costs		Benefits		
		Terminated, Quit, or Reduced Responsibility? (3)	Other Costs (4)	Filed Lawsuit with Potential for Damages (5)	Positive Outcome of Lawsuit (6)	Other Possible Benefits (7)
Heathcouth	Weston Smith, Vice President	Yes	Sentenced to 27 months, forced to pay \$6.9 million.	No		Avoid potential legal liability
ICG	Carl Vogel, CEO	Yes	Left within month after forcing firm to reveal concerns about fraud and accounting.	No		Maintain reputation— within year hired CEO elsewhere
JDN Realty	William Kerley, CFO	No	Alleged loss of \$19 million including legal and job loss costs.	Yes—wrongful dismissal suit	Yes—\$2.3 million	
Northeast Utilities	George Galatin, Engineer	Yes	"If I had it to do over again," says Galatin, "I wouldn't." Alleged by coworkers.	Yes—payment to leave likely	Yes—settlement amount not revealed	On cover of <i>Time</i> magazine

(continued)

(b) Table VIII, part 2.

Employee whistleblowing has high private costs and occasional financial/legal benefits.

Table VIII—Continued

Company (1)	Whistleblower, Position (2)	Costs		Benefits		
		Terminated, Quit, or Reduced Responsibility? (3)	Other Costs (4)	Filed Lawsuit with Potential for Damages (5)	Positive Outcome of Lawsuit (6)	Other Possible Benefits (7)
Oltten	Donald McLendon, executive	Yes	Lost job, could not find other job, alienated from employees.	Yes— <i>qui tam</i>	Yes—\$35 million (significant time delay)	
Quorum	Jim Alderson	Yes	Lost job. Moved to five towns in next 10 years. Left firm a year later.	Yes— <i>qui tam</i>	Yes—\$70 million	
Rite Aid	Joseph Speaker, senior finance executive	No		No		
Service Corporation International Soletron	Charles Albert and Thomas Chaney	Yes		Yes—wrongful dismissal suit	No information	
	Ronald Serishe, division CFO	Yes		Yes—wrongful dismissal suit	No information	Avoid potential legal liability
Xerox	James Bingham, Assistant Treasurer	Yes	"... will never get a job in Corporate America again," according to Bingham's lawyer.	Yes—wrongful dismissal suit	No information	Avoid potential legal liability

(continued)

Table VIII—Continued						
Company (1)	Whistleblower, Position (2)	Costs		Benefits		
		Terminated, Quit, or Reduced Responsibility? (3)	Other Costs (4)	Filed Lawsuit with Potential for Damages (5)	Positive Outcome of Lawsuit (6)	Other Possible Benefits (7)
		Panel B: Unnamed Whistleblowers				
Allegheny Energy	Unnamed executives	Unknown				
America West	Unionized maintenance workers	Unknown				Improve employment conditions
Cendant	Accounting staff integrating newly acquired firm	Unknown				Avoid potential legal liability
Enterways Networks	Unnamed finance executive	Unknown				
Footstar	Corporate Accounting group	Unknown				Avoid potential legal liability
Nisar PlyCor	Anonymous letter Doctors who are employees	Unknown				Improve employment conditions
Symbol Technologies	Unknown letter to SEC	Unknown				
Tenet health care	Unnamed employee	Unknown				
Union Pacific	Union action	Unknown				Improve employment conditions

(a) Table VIII, part 3.

(b) Table VIII, part 4.

Continuation of employee costs and benefits; the case detail motivates the paper's policy emphasis on rewards and protection.

6.6 Table IX: Do Monetary Incentives Impact Employee Whistleblowing?; Table X: Do Regulatory Changes around SOX Affect Whistleblowing?

Read: Table IX tests whether employees whistleblow more in health care, where qui tam monetary incentives can apply. Table X compares detector types before and after SOX.

- Employee whistleblowing is much more common in health care than non-health care.
- The health-care result is not explained away by regulated-industry status.
- Table X shows changes after SOX, including more auditor detections, but no complete restructuring of the fraud-detection ecosystem.

Economic message: Monetary incentives matter for employees. SOX changed some margins but did not replace the need for a broad whistleblower ecosystem.

Do Monetary Incentives Impact Employee Whistleblowing?

This table reports differences in fraud detection between health care industries and non-health care industries. In health care, government purchasing creates the potential for employees to use the *qui tam* statute and derive a monetary benefit from whistleblowing. Panel A reports differences in the distribution of fraud detectors based on our sample of all external whistleblowers. Panel B reports the dismissal rates of suits over our sample period across health care and non-health care industries based on data from Stanford Securities Class Action Clearinghouse. Panel C tests whether employee whistleblowing is more likely in industries where monetary incentives exist as a result of *qui tam* suits, where the dependent variable takes the value of one if the fraud detector is an employee and zero otherwise. Table I provides definitions for the industries included in health care and regulated dummies. The measure of organizational depth is the Rajan and Wulf (2006) measure. *** and ** indicate significant differences at the 1% and 5% levels, respectively.

Panel A: Distribution of Fraud Detectors by Health Care or Other Industries				
	Non-Health Care		Health Care	
	Count	Freq %	Count	Freq %
Analyst	20	14.8%	1	5.9%
Auditor	14	10.4%	2	11.8%
Client or competitor	7	5.2%	—	—
Employee	19	14.1%	7	41.2%
Equity holder	4	3.0%	1	5.9%
Industry regulator	17	12.6%	3	17.7%
Law firm	5	3.7%	—	—
Newspaper	17	12.6%	3	17.7%
SEC	10	7.4%	—	—
Short seller	22	16.3%	—	—
Total	135		17	
Proportions test null hypothesis:				
Proportion (employee whistleblowing, non-health care) – Proportion (employee whistleblowing, health care) = 0				
	Difference	–27.1%	p-Value	0.005
Panel B: Frivolous Suits by Health Care or Other Industries				
	Original Sample	Fraud Cases	Dismissed as Frivolous	Percentage Frivolous
Health care	30	17	13	36.7%
Non-health care	471	199	272	57.8%
Total sample	501	216	285	56.9%
Panel C: Is Employee Whistleblowing More Common in Health Care?				
Logit Estimates: Dependent Variable: Probability of Fraud				

Do Regulatory Changes around the Passage of SOX Affect Whistleblowing?

This table reports differences in the pattern of whistleblowers before and after the passage of the Sarbanes-Oxley Act (SOX) in July of 2002. Columns 2 and 4 report results where observations are weighted by value using the sum of all settlements and fines associated with the class action. ***, **, and * indicate significant differences in the distribution pre- and post-SOX for each category compared to all other categories using a Chi-square distribution test.

	Ended Pre-SOX		Ended Post-SOX	
	Equal Weight (1)	(Value Weight) (2)	(Equal Weight) (3)	(Value Weight) (4)
Analyst	16 (14.0%)	21.3 (17.4%)	5 (13.2%)	2.8 (9.6%)
Auditor	7 (6.1%)	5.9 (4.8%)	9** (23.7%)	5.3*** (18.1%)
Client or competitor	7 (6.1%)	2.7 (2.2%)	—	—
Employee	21 (18.4%)	18.7 (15.2%)	5 (13.2%)	6.9 (23.5%)
Equity holder	4 (3.5%)	4.5 (3.7%)	1* (2.6%)	0.7 (2.4%)
Industry Regulator, Gvt. Agency	13 (11.4%)	10.4 (8.5%)	7 (18.4%)	3.7 (12.6%)
Law firm	5 (4.4%)	3.5 (2.9%)	—	—
Media	17 (14.9%)	31.2 (25.4%)	3 (7.9%)	4.5 (15.4%)
SEC	6 (5.3%)	6.8 (5.5%)	4 (10.5%)	1.8 (6.1%)
Short seller	18 (15.8%)	17.5 (14.3%)	4 (10.5%)	3.7 (12.6%)
Total external governance	114 (100%)	122.7 (100%)	38 (100%)	29.3 (100%)

if we look at the value-weighted number, we do not see this trend. The same
(b) Table X: pre- and post-SOX whistleblowing.

(a) Table IX: monetary incentives in health care.

The policy tables show that incentive design matters, especially for employees.

7 Short summary

- The paper studies 216 large U.S. fraud cases from 1996 to 2004.
- Standard governance actors do not dominate detection.
- Employees, media, industry regulators, analysts, and short sellers are important.
- Access to information and incentives explain who reveals fraud.
- Monetary rewards help explain employee whistleblowing, especially in health care.
- Reputational rewards are weak for analysts but stronger for journalists.
- Employees bear severe costs of whistleblowing, including firing and retaliation.
- SOX changed some detection patterns but did not eliminate the need for a village of whistleblowers.

8 Conclusion

8.1 Core conclusion

Dyck, Morse, and Zingales show that corporate fraud detection is not performed mainly by the actors emphasized in standard governance theories. Instead, detection is decentralized across a village of actors with different information access and incentives.

8.2 Economic intuition

Fraud detection requires both information and incentive. Employees and regulators often have low-cost access to information but weak incentives and high retaliation costs. Short sellers and media may have stronger incentives but need public signals or credible information. Auditors and analysts occupy intermediate positions but face conflicts and weak rewards.

8.3 Incremental takeaway

The paper’s central incremental contribution is to make fraud detection measurable and comparative. It turns a question usually discussed through anecdotes into a taxonomy of detector types, incentives, and costs.

8.4 Policy implications

Effective fraud policy should do more than impose duties on formal gatekeepers. It should protect employees from retaliation, expand rewards where appropriate, preserve media and short-seller channels, and recognize the role of industry regulators with specialized knowledge.

8.5 Limitations

The sample includes revealed frauds, not all frauds. Detector coding is necessarily based on public case histories. The incentive tests are observational. Still, the combined evidence strongly suggests that corporate governance discussions should broaden beyond boards, auditors, SEC enforcement, and investors.

8.6 Takeaway

The paper’s lasting message is that fraud detection is an ecosystem. A robust governance system needs many information holders with enough protection, incentive, and credibility to speak.